Learning Manipulation on HyMeKo Robot Hypergraphs

Demonstrator + Behaviour Cloning + Off-Policy Refinement on Galambos and FANUC

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Abstract

We study whether a structural policy backbone (HSiKAN, signed message passing over the robot's kinematic hypergraph) helps learn two manipulation tasks described in HyMeKo: *Galambos* (two planar arms pull a coin into a target zone) and the *FANUC* LR Mate-config top-down pick-and-place. The central finding is that **pure reinforcement learning cannot crack either task** — it never discovers the long-horizon grasp from a scalar reward — but **behaviour cloning (BC) from a scripted demonstrator breaks this exploration wall**. Beyond BC, on-policy PPO refines modestly while off-policy DDPG/TD3 *degrade* the warm-started policy. The architecture (HSiKAN vs. a params-matched MLP) is effectively tied. FANUC is the tractable task (BC reaches 50% placement); Galambos is bounded by a physical *carry* ceiling ($\sim$ 25%). The evidence points to declared task structure (an FSM / reward machine), not a richer function approximator, as the next lever.

1 The problem: flat RL co-discovers structure and control

A Markov policy on a scalar reward must simultaneously learn the *task decomposition* (reach → grasp → transport → and the *continuous control* within each phase. We measured this failure directly: BC-warm, curriculum PPO on FANUC reached return +93 yet placed 0% greedily; PPO-from-scratch on Galambos stays flat at failure even with the coin spawned *at* the zone (0.071 m from centre). The bottleneck is hard-exploration, not reach, control, backbone, or representation.

2 Behaviour cloning breaks the exploration wall

We script a demonstrator per task — the FANUC IK expert (10/10 grasp), and a Galambos corral→pinch→carry controller built on an analytic planar 2-link IK — collect successful trajectories, and clone the policy's action mean (MSE). BC gives the policy a competent *starting behaviour*:

Task	untrained place	**BC place**	demonstrator ceiling
FANUC pick-and-place	0.00	**0.50**	$\sim$ 1.0 (scripted)
Galambos coin-grasp	0.00	**0.21–0.29**	$\sim$ 0.25 (scripted)

3 Refinement: PPO stable, off-policy degrades

From the BC warm-start we refine. **Galambos, BC→PPO** (HSiKAN vs. params-matched MLP, 3 seeds):

backbone	BC place	→**PPO place** (per seed)	median
HSiKAN	0.04–0.21	0.21, 0.25, 0.29	**0.25**
MLP	0.08	0.17, 0.21, 0.25	**0.21**

HSiKAN edges the MLP (0.25 vs. 0.21) but *within the seed spread* — effectively a tie, both pinned at the control ceiling. **Off-policy (BC→TD3) is unreliable**: HSiKAN 0.083 → 0.125 (slight help), but MLP 0.292 → 0.083 (large degradation). A cold critic pulls the deterministic actor off the BC solution. On FANUC the same pattern appears (BC 0.50; short TD3 refine collapses to 0.0); the full separate DDPG vs. TD3 comparison (× HSiKAN/MLP, 2×10^4 steps each) is running at the time of writing.

4 Why Galambos caps at $\sim 25\%$: a control ceiling

Lowering the coin spawn from the whole table to *at the zone* only moved the return $-224 \rightarrow -198$ (flat, never delivering); fixing a degenerate hand-authored action interface did not change it either. The cause is physical: the coin is a free-spinning cylinder, and a two-finger clamp dragged perpendicular *rolls out*, while the two arms (rooted at $\pm x$) cannot both get behind a central coin to push it. Both RL and a carefully tuned scripted controller top out near 25%. This is the same wall as FANUC's 0% greedy place — a missing structural prior, not a tuning deficit.

5 HyMeKo as the substrate

All of the above rides on HyMeKo descriptions. The robot, observation, reward, and (newly) the whole *scenario* are read from `.hymeko` models, parity-tested against the former Python configuration. The coin and zone were additionally promoted to graph vertices via grasp/goal *hyperedges* (`with_task_hyperedges`), so structure can engage the task objective. A model's composition is itself a hypergraph: each model is a vertex and each model's imports form a composition hyperedge (a hub node), giving a hypergraph-of-hypergraphs (bundle-of-bundles); see `reports/figures/composition.dot`.

6 Conclusion

- Pure RL fails both tasks; **BC is necessary** to break the exploration wall.
- **Architecture is not the lever here**: HSiKAN ≈ MLP at iso-parameters once BC supplies the starting behaviour.
- **Off-policy from a BC warm-start is fragile**; on-policy PPO at least holds the BC level.
- **FANUC is the working learned-control demo** (50% placement from BC); **Galambos is a documented hard case** bounded by physical carry.
- The strongest next lever is **declared task structure** — an FSM / reward machine with HSiKAN leaf controllers — not a richer backbone. This is supported independently by both tasks.

Artifacts: 4 timestamped demonstrator-grasp GIFs (`reports/gifs/demonstrator/`); the model-composition hypergraph (`reports/figures/composition.dot`); checkpoints under `checkpoints/{galambos,fanu`
Companion notes: `reports/2026-06-24-galambos-demonstrator-bc.md`, `2026-06-24-galambos-hyperedge-ab`